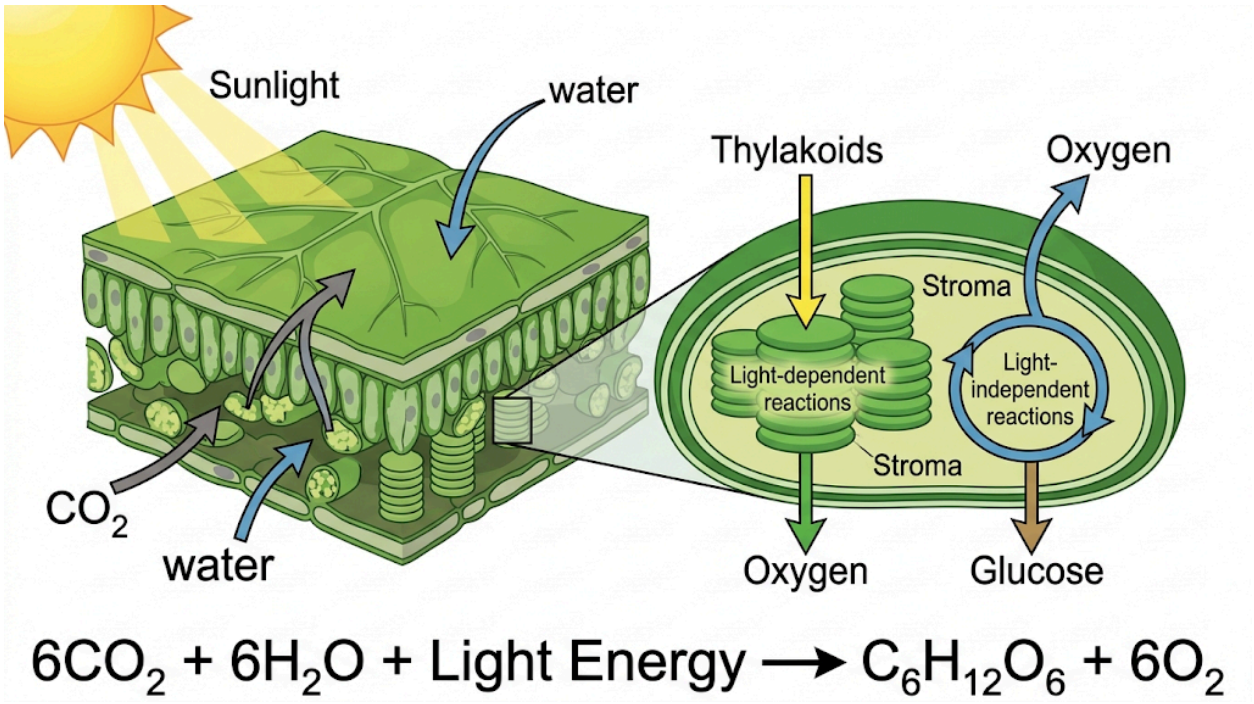
Scenario: I want to make a short paragraph with a picture about how malunggay plants photosynthesize.

Prompt 1; Gemini:
“generate a picture and a short paragraph on how plants photosynthesize”

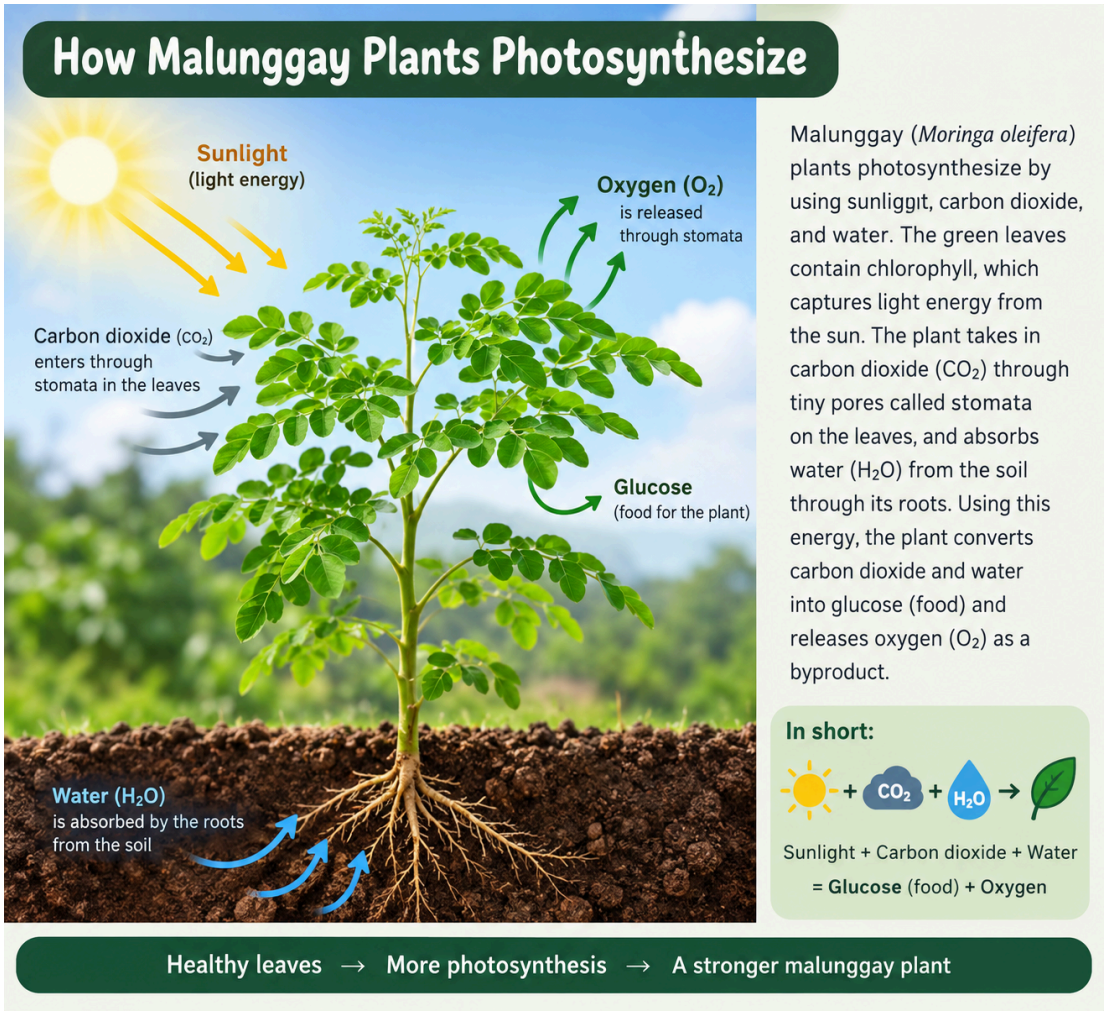
Response:



Photosynthesis is the essential process that plants and other organisms use to convert light energy into chemical energy. As shown in the illustration, this process occurs primarily within the leaves, specifically in organelles called chloroplasts. Chloroplasts contain chlorophyll, the green pigment that captures light energy from the sun. Using this energy, plants take in carbon dioxide (CO2) from the air and water (H2O) from the soil. Through a series of complex reactions, these inputs are transformed into glucose (C6H12O6), a simple sugar that serves as food for the plant, and oxygen (O2), which is released into the atmosphere as a byproduct. This vital process, summarized by the chemical equation 6CO2 + 6H2O + Light Energy -> C6H12O6 + 6O2, forms the foundation of most ecosystems on Earth.

Prompt 2; ChatGPT:
“generate a picture and a short paragraph on how malunggay plants photosynthesize”

Response:



Malunggay (Moringa oleifera) plants make their own food through photosynthesis. Their green leaves contain chlorophyll, which absorbs energy from sunlight. The plant takes in carbon dioxide (CO_2) from the air through tiny pores called stomata and absorbs water (H_2O) from the soil through its roots. Using sunlight, it converts carbon dioxide and water into glucose, which provides energy for growth, and releases oxygen (O_2) into the air.

In simple terms: Sunlight + Carbon dioxide + Water → Glucose + Oxygen

Analysis:
The comparison shows that both Gemini and ChatGPT explain how plants perform photosynthesis by using sunlight, carbon dioxide, and water to produce glucose and oxygen. The difference between the two is that the second prompt specifies what type of plant is being referred to making the response more accurate both in paragraph and in the generated photo. This is a clear example of how proper prompt engineering should work. We must specify our prompts as much as possible in order to get the most desired result, otherwise we might get a subpar/incorrect result.